

Learning Asynchronous Upper-body Task-space Trajectory Tracking Policy for Humanoid Robots

Abstract—High-level humanoid planners often output structurally sparse task-space trajectories at low update rates, whereas whole-body controllers run at high frequency. This creates *temporal asynchrony* between the planning and execution, and *structural incompleteness* for full-body control. We propose an asynchronous upper-body task-space tracking framework for humanoids. A student policy is initialized by teacher–student distillation, conditioned on the full cached future trajectory and an execution-time index, and trained with a sliding-window global reward to learn to handle planning-frame mismatch under asynchronous updates without using the estimated drift for online reference correction. For task-specific post-training, an MPC module completes sparse references into floating-base and upper-body guidance, while action- and FK-level self-guidance constrain policy drift. Simulation and Unitree G1 hardware experiments show improved tracking under low update rates, stronger performance than synchronous and decoupled baselines, and safer adaptation to out-of-distribution motions.

I. INTRODUCTION

Humanoid control has shifted from servo-level tracking to task-level execution. Instead of fixed trajectories or human teleoperation, commands are now generated by high-level planners. This makes the control policy problem an asynchronous upper-body task-space tracking problem. The problem introduces two challenges as shown in Fig. 1. First, the planner runs at a low frequency (1–10Hz), while the low-level controller must operate at a much higher frequency (~ 50 Hz) to maintain stability. Each planner output is executed over multiple control steps. We refer to this planner–controller rate mismatch as *temporal asynchrony*. Second, the reference includes only the end-effector trajectory, the hands and head, in this work. We refer to this structural incompleteness as *structural sparsity*, which leaves the whole-body control problem under-constrained.

Teacher–student distillation is common for control policy learning against structural incompleteness. A teacher policy is trained with a privileged whole-body dataset; a student policy is then distilled from the teacher using only the sparse upper-body reference [1]–[3]. This paradigm works well when references are updated synchronously. Under asynchronous updates, however, the policy experiences a frame mismatch: the reference remains tied to its frame at planning time, while the robot’s frame changes over multiple control steps, as mentioned above. Since the frame drift is not directly observable, existing methods usually rely on two types of remedies. The first explicitly estimates the missing frame transform [4], [5], which breaks end-to-end learning and introduces additional estimation errors. The second converts the sparse reference into local incremental velocity commands [6], [7]. Although this reduces the timing

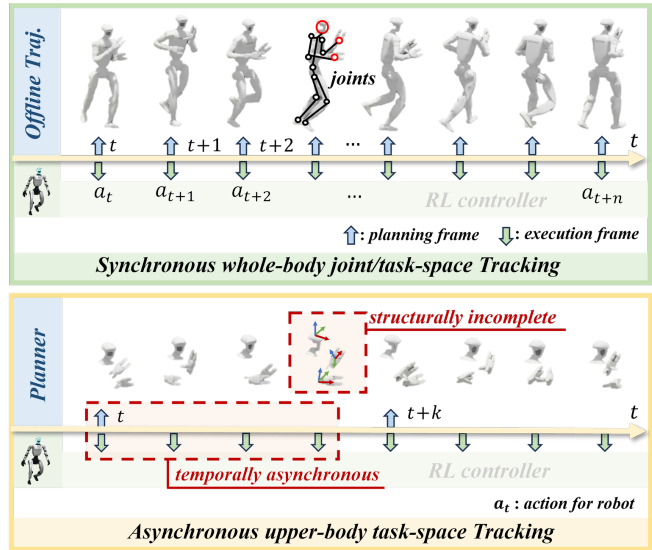


Fig. 1. Execution-interface gap between task-level planners and low-level humanoid controllers. Top: synchronous tracking uses dense, frame-aligned whole-body references. Bottom: low-rate planners provide structurally sparse trajectories, while high-rate controllers execute cached commands. This creates two challenges: (1) *temporal asynchrony*, causing reference-frame mismatch between planning and execution; and (2) *structural incompleteness*, leaving an under-constrained problem for a 29-DoF humanoid.

mismatch, the incremental errors can accumulate over time and lead to severe drift.

When transferring a pre-trained policy to a specific task, zero-shot execution is rarely sufficient. Fine-tuning on task-specific data is typically required [8]. This is easy for whole-body tracking, but becomes challenging when only an upper-body trajectory is available. The structurally incomplete reference complicates task-specific post-training: for whole-body tracking, the same distillation can be applied in both pre-training and post-training. However, when a new task provides only an upper-body trajectory, the privileged teacher, which expects whole-body states, is neither usable nor fine-tunable. The student policy loses its imitation supervision and must rely solely on reinforcement learning (RL) with sparse task rewards. This under-constrains the high-dimensional humanoid policy, leading to unstable training and suboptimal performance.

To address both challenges, we propose a learning framework for asynchronous upper-body trajectory tracking. For temporal asynchrony, we introduce a time-indexed conditioning policy that implicitly aligns the robot’s motion with the reference without explicitly estimating the drift base-frame transform, enabling end-to-end learning. In addition, a sliding-window global reward further penalizes error accumulation over each planning interval. For structural incom-

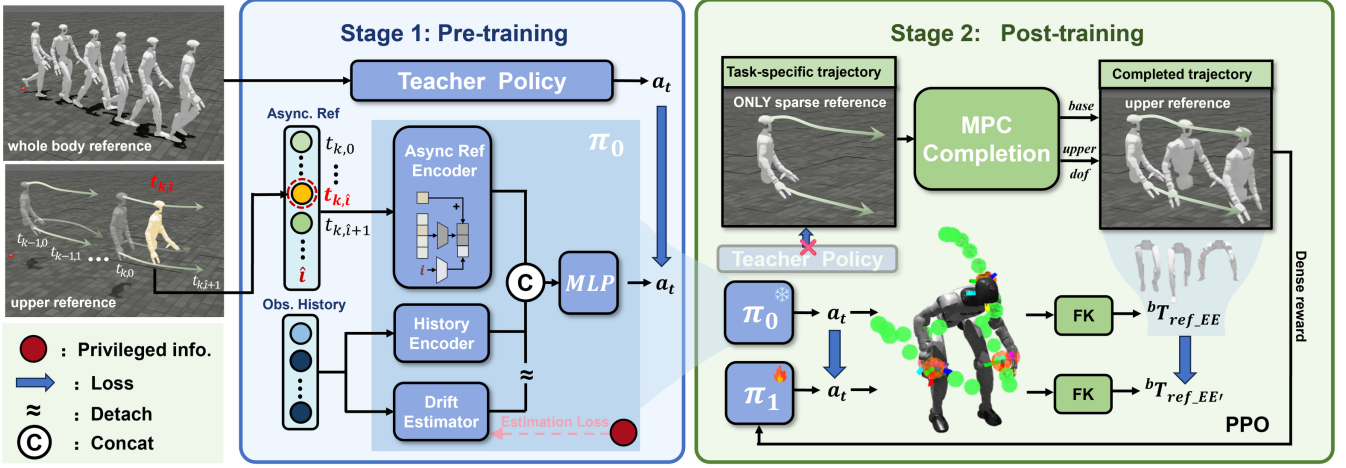


Fig. 2. Overview of the proposed asynchronous sparse trajectory tracking and post-training framework. **Stage 1: Pre-training.** Pretrain the student policy (π_0) via teacher–student distillation with entire sparse references, a time index, history encoding, and auxiliary drift estimation. **Stage 2: Post-training.** Adapts π_0 to task-specific sparse trajectories using MPC-completed base and upper-body references. The pretrained policy π_0 is frozen and serves as a pseudo-teacher, providing action-level and FK-level self-guidance during RL post-training for π_1 .

pleteness, we complete the sparse upper-body trajectory using MPC and incorporate the completed trajectory into both the policy observation and the RL reward. This stabilizes RL post-training. A self-guidance regularization additionally constrains the lower-body behavior, preventing distributional drift. Experiments in the simulation and on a real humanoid robot show that the proposed framework outperforms the baselines and comparative methods. Our contributions are:

- A learning framework that bridges low-frequency, upper-body-only planner outputs to whole-body humanoid control, handling asynchronous and incomplete references.
- A time index conditioned policy with a sliding-window global reward for implicit spatiotemporal alignment.
- A post-training method that integrates MPC-based self-completion with self-guidance regularization.
- Simulation and real-world validation on a humanoid robot, demonstrating stable adaptation to novel motions.

II. RELATED WORKS

A. Task-level Planner Interfaces

Recent humanoid systems are shifting from servo-level tracking to task-level action generation. VLA models and humanoid foundation policies can generate actions from language, vision, proprioception, and demonstrations [6], [9]–[11]. Different planner interfaces provide different levels of embodiment generality and execution guidance. Controller-native latent or action-token interfaces provide rich motion priors but tightly couple the upstream planner to a specific controller and embodiment. For example, GR00T with SONIC predicts latent motion tokens decoded directly by the pretrained whole-body controller [6], [11]. Explicit whole-body interfaces reduce execution ambiguity but remain morphology-specific: VLK predicts root and whole-body joint trajectories, while Ψ_0 predicts upper-body joint actions and structured lower-body commands executed by AMO [10], [12]. In contrast, sparse task-space interfaces

support heterogeneous embodiments and scalable data collection. TrajBooster uses 6D end-effector trajectories as a morphology-agnostic interface, while HuMI obtains task-space keypoint trajectories from robot-free human demonstrations [13], [14]. Such portability leaves whole-body realization more under-specified; our work addresses this setting under low-rate asynchronous planning.

B. Asynchronous Trajectory Tracking

Existing work usually handles the planner–controller gap by making low-rate task commands executable for a high-rate controller. A common approach is to buffer a future action chunk and execute the waypoint that matches the current timestep [6], [14], [15]. This resolves the rate mismatch between planning and control. In this case, the selected waypoint is generated in the planning-time frame, but is executed as an instantaneous target under the current floating-base state, leaving the geometric mismatch unresolved. Another line of work use local incremental commands, such as base velocities or short-horizon relative keypoint motions [7], [9], [10], [14]. They compress a future trajectory into step-wise local updates, so the long-horizon positional structure is weakened and integration errors may accumulate over time. More geometric treatments reconstruct or refresh the tracking target during execution. For example, HiWET formulates end-effector tracking in a world frame to maintain spatial consistency [4], while HERO improves tracking with learned FK/odometry, goal adjustment, and periodic replanning [5]. These methods reduce geometric drift, but they usually rely on global localization, state estimation, or online correction. Also, their odometry always assumes planted feet, which fails under the moving-base trajectories considered in this work. The estimator and controller are also often optimized separately. In contrast, our policy observes the entire future trajectory segment together with an execution-phase index. This allows the controller to learn the consistency between the planning-time reference and the current state in an end-to-end manner without explicit frame estimation.

C. Tracking Policy Post-training

Recent humanoid control works show that pretrained whole-body controllers often require post-training when the task or reference distribution changes [16]–[18]. LIFT studies efficient fine-tuning of pretrained humanoid policies in new environments and out-of-distribution (OOD) tasks [8]. AHC further follows a distillation-then-adaptation pipeline, where a multi-behavior controller is first obtained by behavior distillation and then improved through reinforced fine-tuning [19]. However, these methods do not directly address controller post-training from low-rate upper-body-only structurally sparse trajectories, where the floating-base and lower-body motions are unspecified. In teacher–student trackers, the dense teacher supervision is unavailable when adapting to new sparse references and direct RL fine-tuning with sparse rewards is also prone to policy drift or forgetting [20], [21]. We therefore use self-guided post-training, where the pretrained student acts as a pseudo-teacher and MPC-completed references provide auxiliary rewards.

III. OVERVIEW

We propose a two-stage framework to address asynchronous upper-body task-space tracking, as illustrated in Fig. 2. In Stage 1 (Sec. IV), the pre-training stage, a whole-body motion dataset is available. We adopt teacher–student distillation and introduce ASYNC-3PT to handle temporal asynchrony. In Stage 2 (Sec. V), the post-training stage, only task-specific upper-body trajectories are provided. We therefore extend ASYNC-3PT to ASYNC-CA by additionally considering the base motion beyond the head and hands. To this end, we employ MPC to complete the base trajectory given the upper-body reference, which augments the policy observation and enables self-adaptation under structural incompleteness. Together, the two stages form a complete framework for the asynchronous tracking problem.

IV. ASYNC-3PT: TIME-INDEXED CONDITIONING FOR ASYNCHRONOUS TRACKING

This section formalizes Stage 1 of our framework. We first define and derive the frame mismatch caused by asynchronous execution, and then formulate the ASYNC-3PT policy to consider the asynchronous time.

We assume that, a floating-base humanoid is commanded by a high-level planner whose update rate is lower than the control frequency. At each planning time t_k , the planner produces a finite-horizon reference for task-relevant end-effectors, such as the head and dual hands (Fig. 3, left). Let $t_{k,i} = t_k + i\tau$ denote the timestamp of the i -th future waypoint, where $i = 0, \dots, N_{\text{ref}} - 1$ and τ is the waypoint spacing. The planner provides the relative end-effector motion ${}^{ee_{t_k}}\mathbf{T}_{ee_{t_{k,i}}}^{\text{ref}} \in SE(3)$, which describes the future pose change of the end-effector with respect to its pose at the planning time t_k . Using the end-effector pose from forward kinematics,

$${}^{b_{t_k}}\mathbf{T}_{ee_{t_k}} = \text{FK}(\mathbf{q}_{t_k}), \quad (1)$$

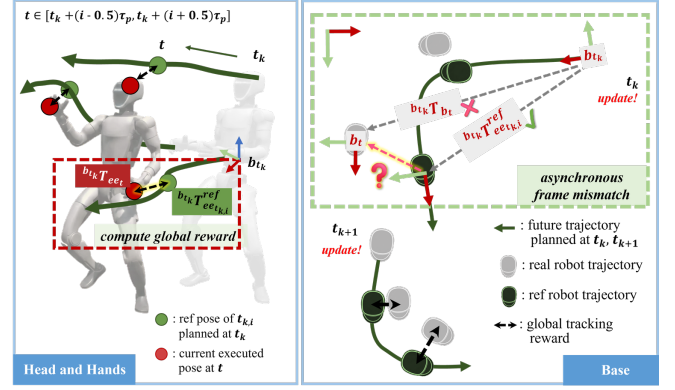


Fig. 3. **Asynchronous frame mismatch and sliding-window global reward.** **Base (right):** the base drifts from b_{t_k} to b_t between updates, so the transform ${}^{b_{t_k}}\mathbf{T}_{b_t}$ is unavailable, causing the mismatch of the reference frame. **Head and Hands (left):** we keep the cached reference and executed pose in the same frame b_{t_k} , and penalize their global discrepancy over a sliding window.

where $\mathbf{q}_{t_k}$ denotes the joint positions, we express the sparse future trajectory in the planning-time base frame as

$${}^{b_{t_k}}\mathbf{T}_{ee_{t_{k,i}}}^{\text{ref}} = {}^{b_{t_k}}\mathbf{T}_{ee_{t_k}} {}^{ee_{t_k}}\mathbf{T}_{ee_{t_{k,i}}}^{\text{ref}}. \quad (2)$$

The sparse reference is therefore

$$\mathcal{T}_k = \left\{ {}^{b_{t_k}}\mathbf{T}_{ee_{t_{k,i}}}^{\text{ref}} \right\}_{i=0}^{N_{\text{ref}}-1}. \quad (3)$$

It is a low-rate, structurally sparse reference defined in the planning-time base frame b_{t_k} , covering only a subset of end-effectors.

Unless otherwise specified, each cached reference $\mathcal{T}_k$ contains $N_{\text{ref}} = 20$ future waypoints with spacing $\tau = 0.1$ s. The controller operates at $\Delta t_c = 0.02$ s (50 Hz). We use $\Delta t_k = 1$ s during training.

A. Asynchronous Execution and Frame Mismatch

The low-level controller runs at a higher frequency than the planner. As a result, each trajectory $\mathcal{T}_k$ is tracked over multiple control steps until the next reference update at t_{k+1} . During this interval, the floating base keeps moving, while the reference trajectory remains defined in the planning-time base frame b_{t_k} . At execution time $t > t_k$, the robot observations are expressed in the current base frame b_t . Ideally, one should transform the planned reference from b_{t_k} to b_t :

$${}^{b_t}\mathbf{T}_{ee_{t_{k,i}}}^{\text{ref}} = {}^{b_t}\mathbf{T}_{b_{t_k}} {}^{b_{t_k}}\mathbf{T}_{ee_{t_{k,i}}}^{\text{ref}}. \quad (4)$$

However, the relative transform ${}^{b_t}\mathbf{T}_{b_{t_k}}$ is not directly available without reliable global localization, leading to mismatch of the reference frame. As a result, the planned reference and the current robot state are represented in different frames, so the task-space tracking error cannot be directly measured, making asynchronous tracking challenging.

B. Asynchronous Time-indexed Conditioning

To handle this mismatch in an end-to-end manner, we introduce a sparse end-effector asynchronous controller, denoted as ASYNC-3PT. The key idea is to condition the policy

on the entire future trajectory segment and a time index that indicates the current execution step against the planning step. Rather than relying only on a single waypoint, the policy observes the entire sparse trajectory and learns to infer the proper alignment implicitly.

1) *Policy Formulation*: At control step t , the policy receives

$$o_t = [o_t^{\text{prop}}, \mathbf{h}_t, \hat{i}, o_t^{\text{task}}], \quad (5)$$

and outputs PD joint targets $a_t \in \mathbb{R}^{29}$. The proprioceptive observation is defined as $o_t^{\text{prop}} \triangleq [\mathbf{q}_t, \dot{\mathbf{q}}_t, \boldsymbol{\omega}_t, \mathbf{g}_t]$, including joint positions, joint velocities, base angular velocity, and projected gravity. The history buffer is defined as $\mathbf{h}_t \triangleq [o_{t-L_h:t-1}^{\text{prop}}, \mathbf{a}_{t-L_h:t-1}]$ with $L_h = 25$. Given the time elapsed since the last planning step is $t - t_k$, we define time index as

$$\hat{i} = \left\lfloor \frac{t - t_k}{\tau} \right\rfloor. \quad (6)$$

The policy retains the entire cached trajectory and additionally receives a phase-aligned reference token, preserving both future motion context and the current execution phase. The task observation is defined as $o_t^{\text{task}} \triangleq [\mathcal{T}_k, \Delta\mathcal{T}_k]$, where $\Delta\mathcal{T}_k$ is the local SE(3) errors between the reference and the current end-effector and head poses at the update time t_k . The time index $\hat{i}$ and o_t^{task} jointly guide the robot to capture the asynchrony and motion context.

2) *Auxiliary Drift Estimator*: To further improve robustness under asynchronous reference-frame mismatch, we introduce an auxiliary head to estimate the base height and each end-effector's pose change relative to its pose at the latest update time t_k . The estimator takes the features from the observation history, and employs a lightweight decoder for prediction. Note that the predicted frame drift is not applied to transform the reference trajectory, but acts as an auxiliary supervision signal for regularization.

C. Teacher-Student Distillation

We train the policy via teacher-student distillation, as shown in Fig. 2 (Stage 1). The teacher policy is pre-trained on a large-scale whole-body motion dataset with access to privileged information (e.g., globally-consistent state signals), while the student only observes the asynchronous input defined in Sec. IV-B. The loss for the student policy includes both the imitation and the reinforcement learning:

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{imit}} + \mathcal{L}_{\text{PPO}}. \quad (7)$$

1) *Imitation Learning*: The imitation objective contains action distillation and auxiliary state estimation:

$$\mathcal{L}_{\text{imit}} = \lambda_{\text{act}} \mathcal{L}_{\text{act}} + \lambda_{\text{se}} \mathcal{L}_{\text{se}}. \quad (8)$$

The action distillation term is defined as

$$\mathcal{L}_{\text{act}} = \|\mathbf{a}_t^{\text{stu}} - \mathbf{a}_t^{\text{tea}}\|_2^2, \quad (9)$$

where $\mathbf{a}_t^{\text{tea}}$ and $\mathbf{a}_t^{\text{stu}}$ are the teacher and student actions, respectively.

The auxiliary estimation loss is

$$\mathcal{L}_{\text{se}} = \mathcal{L}_{\text{ee}}(\Delta\hat{\mathbf{p}}_t, \Delta\hat{\mathbf{R}}_t, \Delta\mathbf{p}_t, \Delta\mathbf{R}_t) + \mathcal{L}_{\text{height}}(\hat{h}_t, h_t), \quad (10)$$

where, for each end effector $e \in \{\text{lh}, \text{rh}, \text{head}\}$,

$$\Delta\mathbf{p}_t^e = (\mathbf{R}_{t_k}^e)^\top (\mathbf{p}_t^e - \mathbf{p}_{t_k}^e), \quad \Delta\mathbf{R}_t^e = (\mathbf{R}_{t_k}^e)^\top \mathbf{R}_t^e.$$

Here, $(\mathbf{p}_t^e, \mathbf{R}_t^e)$ is the executed EE pose in a common world frame; its relative change is expressed in the EE frame at t_k . We supervise the relative position, 6D rotation representation, and current base height using simulator ground truth available only during training.

2) *Reinforcement Learning*: Since the teacher policy has access to the global observation and reference, its action guidance may cause the ambiguity for the student. So we employ the reinforcement learning for student policy.

The PPO loss $\mathcal{L}_{\text{PPO}}$ is optimized with motion regularization and tracking reward. Importantly, we design a sliding-window global tracking reward to improve global position tracking within a re-planning interval. At time t , we evaluate tracking in the latest planning-time base frame b_{t_k} , as shown in Fig. 3. Specifically, both the reference and executed end-effector poses are expressed in b_{t_k} , and the SE(3) tracking error is

$$\text{Log} \left(\left({}^{b_{t_k}}\mathbf{T}_{ee}^{\text{ref}}(t) \right)^{-1} {}^{b_{t_k}}\mathbf{T}_{ee_t} \right), \quad (11)$$

The reference at t is interpolated at the controller rate. Here, $\text{Log}(\cdot)$ maps the relative transform to a 6D error in $\mathfrak{se}(3)$. This reward is evaluated at every control step and resets when a new planner command arrives, thus penalizing drift accumulated within each planning window.

V. ASYNC-CA: COMPLETION-AUGMENTED FOR POST-TRAINING

ASYNC-3PT tracks sparse head-and-hand task-space references without explicit frame alignment. However, direct RL post-training of this controller is not effective. This is because the task-specific data contain only sparse trajectories, making the standard teacher-student distillation pipeline inapplicable. To overcome this, we propose an MPC-based completion module to complete floating-base and upper-body joint trajectories from the sparse reference, and augment the ASYNC-3PT observation and reward, resulting in the controller ASYNC-CA. We pretrain ASYNC-CA (π_0 in Fig. 2) using the same teacher-student distillation as Sec IV-C. During post-training, the MPC-completed references provide augmented information for adaptation. Combined with two-level self-guidance, this stabilizes post-training under sparse references, as shown in Stage 2 in Fig. 2.

A. MPC-based Trajectory Completion and Policy Augmentation

The high-level planner provides sparse task-space references $\mathcal{T}_k$ only for the head and two hands, leaving the floating-base motion and upper-body joint configurations unspecified. We therefore use a kinematic MPC to complete these sparse references into constraint-aware base and upper-body trajectories, which serve as dense guidance for

TABLE I
COMPARISON OF DIFFERENT CONTROLLER DESIGNS UNDER SYNCHRONOUS AND ASYNCHRONOUS EXECUTION SETTINGS

Method	Succ. (%) $\uparrow$	Pos _{1s} $\downarrow$	Rot _{1s} $\downarrow$	Pos _{loc} $\downarrow$	Rot _{loc} $\downarrow$	LinVel _{1s} $\downarrow$	AngVel _{1s} $\downarrow$	LinVel _{loc} $\downarrow$	AngVel _{loc} $\downarrow$
TEACHER	99.32	/	/	4.08 $\pm$ 1.38	10.06 $\pm$ 2.82	/	/	0.45 $\pm$ 0.17	0.93 $\pm$ 0.17
Sync W.SC	98.93	/	/	4.25 $\pm$ 1.46	10.43 $\pm$ 2.79	/	/	0.46 $\pm$ 0.17	0.90 $\pm$ 0.18
Sync W.ASC	89.94	7.86 $\pm$ 3.89	13.79 $\pm$ 5.31	6.16 $\pm$ 2.62	12.37 $\pm$ 4.05	0.53 $\pm$ 0.32	0.99 $\pm$ 0.38	0.51 $\pm$ 0.19	1.07 $\pm$ 0.29
Sync W.explicit estimator	62.16	6.55 $\pm$ 1.56	9.15 $\pm$ 2.58	6.39 $\pm$ 1.67	9.37 $\pm$ 2.80	0.47 $\pm$ 0.14	0.80 $\pm$ 0.14	0.47 $\pm$ 0.13	0.99 $\pm$ 0.23
ASYN-3PT from scratch	73.63	12.46 $\pm$ 3.49	29.44 $\pm$ 6.79	11.51 $\pm$ 3.04	27.85 $\pm$ 6.96	0.61 $\pm$ 0.24	1.46 $\pm$ 0.41	0.62 $\pm$ 0.24	1.54 $\pm$ 0.40
ASYN-3PT (ours)	98.93	5.91 $\pm$ 1.53	11.53 $\pm$ 2.88	4.55 $\pm$ 1.48	10.82 $\pm$ 2.96	0.49 $\pm$ 0.15	0.74 $\pm$ 0.10	0.47 $\pm$ 0.15	0.90 $\pm$ 0.19
Decoupled	94.336	11.804 $\pm$ 3.733	17.678 $\pm$ 6.331	3.223 $\pm$ 1.870	11.649 $\pm$ 5.353	0.497 $\pm$ 0.192	1.920 $\pm$ 0.917	0.416 $\pm$ 0.188	1.676 $\pm$ 0.871
ASYN-CA w.o. estimator	98.87	5.54 $\pm$ 1.74	11.36 $\pm$ 2.90	4.35 $\pm$ 1.58	10.56 $\pm$ 3.02	0.48 $\pm$ 0.18	0.72 $\pm$ 0.11	0.47 $\pm$ 0.17	0.87 $\pm$ 0.21
ASYN-CA (ours)	99.17	4.87 $\pm$ 1.46	10.94 $\pm$ 2.77	4.08 $\pm$ 1.44	10.15 $\pm$ 2.91	0.47 $\pm$ 0.17	0.76 $\pm$ 0.12	0.46 $\pm$ 0.16	0.90 $\pm$ 0.21

policy execution and post-training. We extend the original reference trajectory in (3) with the MPC-completed floating-base trajectory and augment o_t^{task} in (5) with the completed upper-body joint references. We further introduce upper-body tracking rewards and an asynchronous sliding-window reward for base tracking. Together, these observation and reward augmentations provide denser supervision for post-training under sparse planner commands.

B. Self-guidance Reinforcement Learning

In post-training, we use the MPC-completed base and upper-body references. These completed references provide additional upper-body and base guidance. However, the lower body is still not directly specified by the sparse task-space reference. Therefore, direct RL adaptation may satisfy the sparse tracking objective while producing abnormal lower-body motions. To preserve the pretrained motion prior, we introduce two-level self-guidance regularization during post-training, as indicated by the blue arrows in Fig. 2(right).

We use the pretrained policy π_0 as a pseudo-teacher for the adapted policy π_1 . Let $\mathbf{a}_t^1 = \pi_1(\mathbf{o}_t)$ and $\mathbf{a}_t^0 = \pi_0(\mathbf{o}_t)$ denote their predicted actions. The action-level self-guidance is defined as

$$\mathcal{L}_{\text{sd}}^{\text{act}} = \mathbb{E} \left[\left\| \mathbf{a}_t^1 - \mathbf{a}_t^0 \right\|_2^2 \right]. \quad (12)$$

This term constrains the adapted policy to stay close to the pretrained action prior, reducing jerky action variations. We further apply FK-level self-guidance on selected body poses:

$$\mathcal{L}_{\text{sd}}^{\text{fk}} = \mathbb{E} \left[\left\| \text{FK}(q^{\text{tar}}(\mathbf{a}_t^1)) - \text{FK}(q^{\text{tar}}(\mathbf{a}_t^0)) \right\|_2^2 \right], \quad (13)$$

where $\text{FK}(\cdot)$ maps the decoded target joint configuration to selected body poses. This term regularizes the adapted motion in Cartesian space. In practice, we upweight lower-body poses in the FK-level loss because they are not directly constrained by sparse head-and-hand tracking and are more prone to drift.

Together, the dense rewards from MPC-completed references and the two-level self-guidance regularization allow the policy to adapt to novel tasks while avoiding catastrophic drift from the pretrained motion prior.

VI. EXPERIMENTAL RESULTS

We evaluate the proposed framework along two axes. First, we study whether asynchronous modeling is necessary for

sparse trajectory tracking under low-frequency references, and analyze tracking performance across different update frequencies. Second, we examine whether MPC-completed guidance improves post-training beyond sparse 3-point guidance on out-of-distribution tasks, and evaluate the effect of self-guidance in both simulation and real-robot experiments.

A. Experimental Setup and Metrics

We evaluate the proposed framework on the 29-DoF Uni-tree G1 humanoid in simulation and on the real robot. The controller receives sparse future trajectories with a horizon of 20 frames and runs at 50 Hz. Unless otherwise specified, the sparse references are updated at 1 Hz, corresponding to a 1 s reference interval.

We report success rate and tracking errors. Following SONIC [11], *Succ.* counts a rollout as successful if it completes without falling or exceeding a vertical tracking error of 0.25 m at any tracked end effector. Our primary metrics are the asynchronous interval errors: Pos_{1s} and Rot_{1s}, which are computed over one asynchronous reference interval and measure frame drift from the latest planning time using the same planning-frame error definition as Eq. (11). The local metrics Pos_{loc} and Rot_{loc} are computed in the root-relative frame and measure local end-effector tracking accuracy. All position errors are reported in centimeters, rotation errors in degrees, linear velocity errors in m/s, and angular velocity errors in rad/s. Table I reports statistics over all rollouts, including failures.

B. Effect of Asynchronous Sparse Trajectory Tracking

We first evaluate whether explicit asynchronous modeling is necessary for executing low-frequency sparse references. We consider the following controller variants:

- **TEACHER**: a privileged teacher policy adopted from the OmniH2O framework [22].
- **Sync W.SC**: a synchronous sparse tracker trained in the OmniH2O frame [22], evaluated with synchronized references and the same future horizon.
- **Sync W.ASC**: the same synchronous tracker is directly deployed under asynchronous reference updates, distilled from TEACHER.
- **ASYN-3PT from scratch**: ASYN-3PT trained directly from scratch with RL. This baseline examines

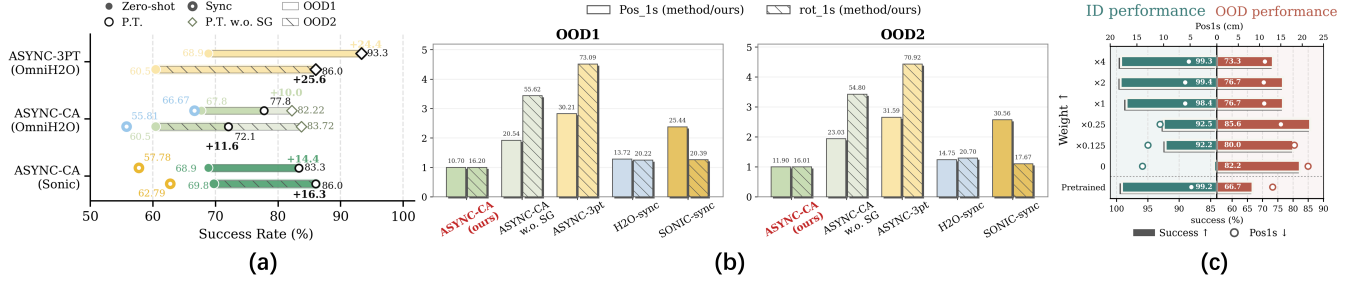


Fig. 4. **OOD Adaptation under Sparse Task-Space References.** (a) Success rates on *ood1* and held-out *ood2*. Filled markers denote zero-shot deployment and hollow markers denote post-training (P.T.). Half-filled markers denote the original synchronous policy under asynchronous updates, which serves as a baseline. ASYNC-3PT uses only three-point end-effector references; ASYNC-CA uses MPC-completed base and upper-body guidance with self-guidance. (b) Interval tracking errors on *ood1* and *ood2*. Each bar reports the ratio of Pos_{1s} or Rot_{1s} to ASYNC-CA (ours), where ASYNC-CA is normalized to 1.0; larger values indicate larger asynchronous drift. (c) Comparison of different self-guidance weights. Bars show the absolute success rate and circles show Pos_{1s}.

whether sparse head-and-hand references alone are sufficient for asynchronous tracking.

- **Sync W.explicit estimator:** a synchronous tracker that explicitly corrects cached references using learned drift estimates.

Since both TEACHER and Sync W.SC use unavailable information, they are reported only as references. The remaining controllers are evaluated under asynchronous reference updates. All methods are trained and evaluated using the motion dataset from OMOMO and GRAB [23], [24].

Effect of asynchronous modeling. As shown in Table I, ASYNC-3PT achieves 98.93% success and reduces the interval position error from 7.86 cm for Sync W.ASC to 5.91 cm, demonstrating more reliable asynchronous execution.

TABLE II
Ablation of time-indexed conditioning on a held-out trajectory set.

Time index	Succ. ↑	Pos _{1s} ↓	Rot _{1s} ↓	Vel _{1s} ↓	AngVel _{1s} ↓
✗	98.9	14.6	26.6	0.550	1.298
✓	99.4	8.6	12.8	0.416	1.048

Effect of time-indexed conditioning. We further ablate the time-indexed conditioning on a separate unseen trajectory set. As shown in Table II, removing the time index only slightly affects success rate, but increases Pos_{1s} from 8.6 cm to 14.6 cm and Rot_{1s} from 12.8° to 26.6°. This confirms that the time index mainly helps resolve the phase mismatch within each asynchronous reference interval. The time index also enables inference-time speed modulation, while higher re-planning frequencies reduce tracking errors [25].

C. Comparison of Base-augmented Tracking

We compare the last two controller variants in Table I. The Decoupled baseline uses the same base and upper-body references and observations as ASYNC-CA, but separates base tracking and upper-body tracking into two different control modules. This design is related to recent decoupled tracking frameworks such as HoldMyBeer [26] and PMP [27]. Although Decoupled achieves lower local position error, ASYNC-CA reduces asynchronous errors. Thus, better root-relative accuracy does not imply better asynchronous tracking.

We also ablate the state estimator in ASYNC-CA. Removing the auxiliary estimator slightly reduces success and increases interval position/rotation errors. This indicates a modest pose-tracking benefit from auxiliary supervision, without requiring its predictions for online reference correction.

D. Closed-loop Online Manipulation

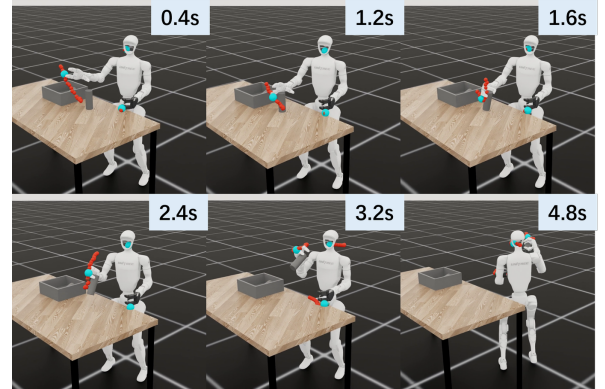


Fig. 6. GR00T+ASYNC-CA cup-grasping execution with online planner.

TABLE III
ONLINE CLOSED-LOOP TASK COMPARISON.

Method	Planner interface	Success ↑	Tracking error ↓
GR00T + ASYNC-CA	Sparse: 3-EE traj.	12/25	6.69 ±0.11 / 8.41 ±0.58
Ψ ₀ + AMO	Dense: base vel. + upper-body joints	6/25	7.23±0.44 / 9.29±0.67

To measure closed-loop tracking errors with a real-time planner, we integrate ASYNC-CA with a GR00T-based planner [6], which predicts head-and-hand trajectories from the current observation every 0.8 s. Online MPC completes them into base and upper-body references. We also train a Ψ₀+AMO baseline that predicts dense upper-body and base action chunks. Both planners use the same 200 simulated demonstrations collected with VIRAL [28] and are evaluated on 25 cup-grasping trials from random initial positions. Table III reports success rates and asynchronous position and rotation tracking errors. Our pipeline achieves higher success (48% vs. 24%) and lower tracking errors, while

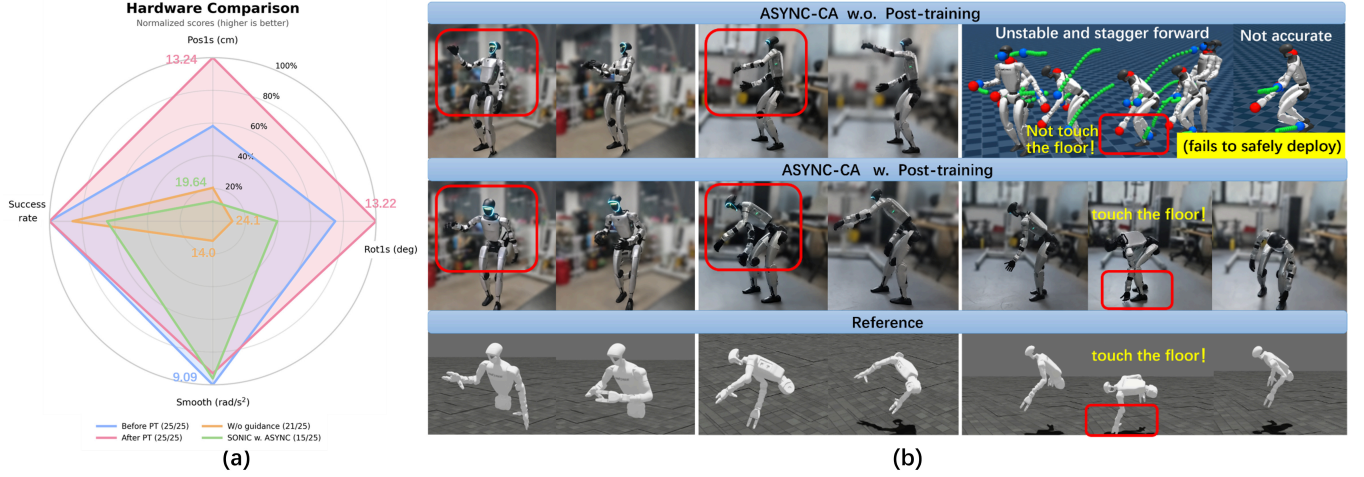


Fig. 5. Hardware comparison and qualitative post-training evaluation. (a) Five-motion hardware comparison (25 trials per policy). Each metric is reverse min–max normalized across the four policies (higher is better); radial percentages denote normalized scores, and legend counts denote successful trials. (b) Without post-training, **ASYNC-CA** exhibits inaccurate tracking and unstable forward staggering, where the reference motion is nearly in-place. With post-training, the policy better follows the reference and maintains a stable whole-body posture in both simulation and real-world execution. Red boxes highlight representative differences.

requiring the planner to predict only three sparse end-effector trajectories rather than dense base-and-upper-body action chunks. Since the two planners use different interfaces and horizons, this comparison evaluates the complete online pipelines rather than isolating the controller alone. Under identical Ψ_0 predictions at 0.46-s intervals, **ASYNC-CA** also yields lower and more stable tracking errors than **AMO** [25].

E. OOD Post-Training with MPC Completion

We construct a failure-mined hard-motion OOD benchmark from **AMASS** by zero-shot evaluating the **OmniH2O** teacher and selecting 120 trajectories on which it fails or incurs large tracking errors. These trajectories do not overlap with the pretraining or standard evaluation data. We use 80 motions for post-training (*ood1*) and hold out 40 for evaluation (*ood2*). Using **OmniH2O** [22] and **SONIC** [11] as privileged teacher backbones, we compare **ASYNC-3PT**, post-trained via RL alone on the original sparse head-and-hand references, with **ASYNC-CA**, post-trained using MPC-completed base and upper-body guidance with and without self-guidance (Fig. 4).

Effect of MPC-completed guidance. RL-only post-training of **ASYNC-3PT** achieves higher success under the **OmniH2O** backbone but leaves large asynchronous tracking errors, especially rotational drift. Combining MPC-completed base and upper-body guidance with self-guidance, **ASYNC-CA** reduces the *ood1* position/rotation errors from 30.21 cm/73.09° to 10.70 cm/16.20°, with similar reductions on held-out *ood2* (Fig. 4(b)). After post-training, **ASYNC-CA** also achieves higher success and lower interval errors than synchronous controllers deployed under asynchronous reference updates (Fig. 4(a,b)), supporting the effectiveness of the combined guidance for OOD adaptation.

Effect of self-guidance. We jointly scale the action- and FK-level self-guidance weights by $w \in \{0, 0.125, 0.25, 1, 2, 4\}$ and evaluate ID retention on **OMOMO+GRAB** and adaptation on *ood1*. Figure 4(c)

reveals a clear adaptation–preservation trade-off. Without self-guidance, post-training improves *ood1* success by 15.5 percentage points, but ID success drops from 99.2% to 84.5%, while the OOD position error increases to 21.36 cm. In contrast, $w = 2$ preserves ID success while still improving OOD success by 10.0 percentage points and reducing the position error to 10.96 cm. Overall, stronger self-guidance better preserves the pretrained ID behavior, whereas weaker guidance permits greater OOD adaptation at the cost of increased policy drift.

Effect of teacher backbone. With the **SONIC** teacher, **ASYNC-CA** reaches 83.3% success on *ood1* and 86.0% on *ood2*. The consistent improvement obtained with the same post-training procedure indicates that the method is not tied to a particular teacher backbone.

F. Real-world Hardware Experiments

Quantitative hardware evaluation. We deploy 14 motions spanning walking, bending/squatting, kneeling, standing up, and manipulation on **G1**, with 60/70 trials completed without falling (Table IV). On the five motions selected by zero-shot simulation success, P.T. improves tracking while retaining 25/25 success (Fig. 5(a)). Smooth is the temporal mean of the joint-wise RMS acceleration of 17 measured upper-body joints (lower is better). Removing self-guidance reduces success to 21/25 and increases Smooth from 9.476 to 14.008 rad/s², showing that self-guidance improves both robustness and motion smoothness.

Qualitative self-guidance evaluation. Without self-guidance, the underconstrained lower body develops unrealistic motions such as foot eversion; self-guidance preserves natural lower-body behavior and improves real-world transfer (Fig. 7).

VII. CONCLUSION

This letter presented an asynchronous upper-body task-space trajectory tracking policy for humanoid robots, using

TABLE IV

SIMULATION AND HARDWARE PERFORMANCE ON
14 SELECTED MOTIONS FOR REAL-WORLD DEPLOYMENT. SIMULATION
METRICS USE ALL-MOTION STATISTICS.

Eval.	Policy	Succ.	Pos _{1s} ↓ (cm)	Rot _{1s} ↓ (°)	Smooth ↓ (rad/s ²)
Simulation	Zero-shot	5/14 (35.7%)	12.455	13.270	13.35
	w/o SG	6/14 (42.9%)	12.727	13.592	14.465
	SONIC-w. ASY	6/14 (42.9%)	28.095	18.129	11.810
	After PT	12/14 (85.7%)	11.273	12.147	11.50
Hardware	After PT	60/70 (85.7%)	13.42	14.32	10.52

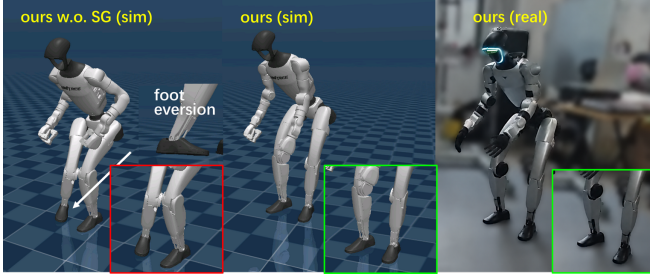


Fig. 7. We compare post-trained policies with and without self-guidance(SG). Removing self-guidance leads to unrealistic lower-body motions, such as foot eversion. Self-guidance regularizes the lower body and improves real-world transfer.

time-indexed conditioning and sliding-window rewards for handling of asynchronous planning-frame mismatch, without online explicit reference reprojection, plus MPC-based completion and self-guidance for stable RL adaptation. Simulation and hardware validation on the G1 confirms effectiveness and improvements. Future work may investigate the end-to-end training of both planner and controller.

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